

Mining Greek Job Postings: Estimating Role Frequency & Regional Variation

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Thesis: MSc in Mobile and Web
Computing



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The Possibilities of Utilizing Existing Data

- Current job portals only offer a limited perspective.
- Deriving to insights by aggregating the real time data of job portals is feasible and relatively effective.
- These insights could be useful for a plethora of stakeholders and not only for employees.

Scope of study

- This Thesis main objective was to develop methods for measuring the parameters of a certain field and the relations between them through the job postings.
- To track technological adoption directly from active demands.

Implementation Overview

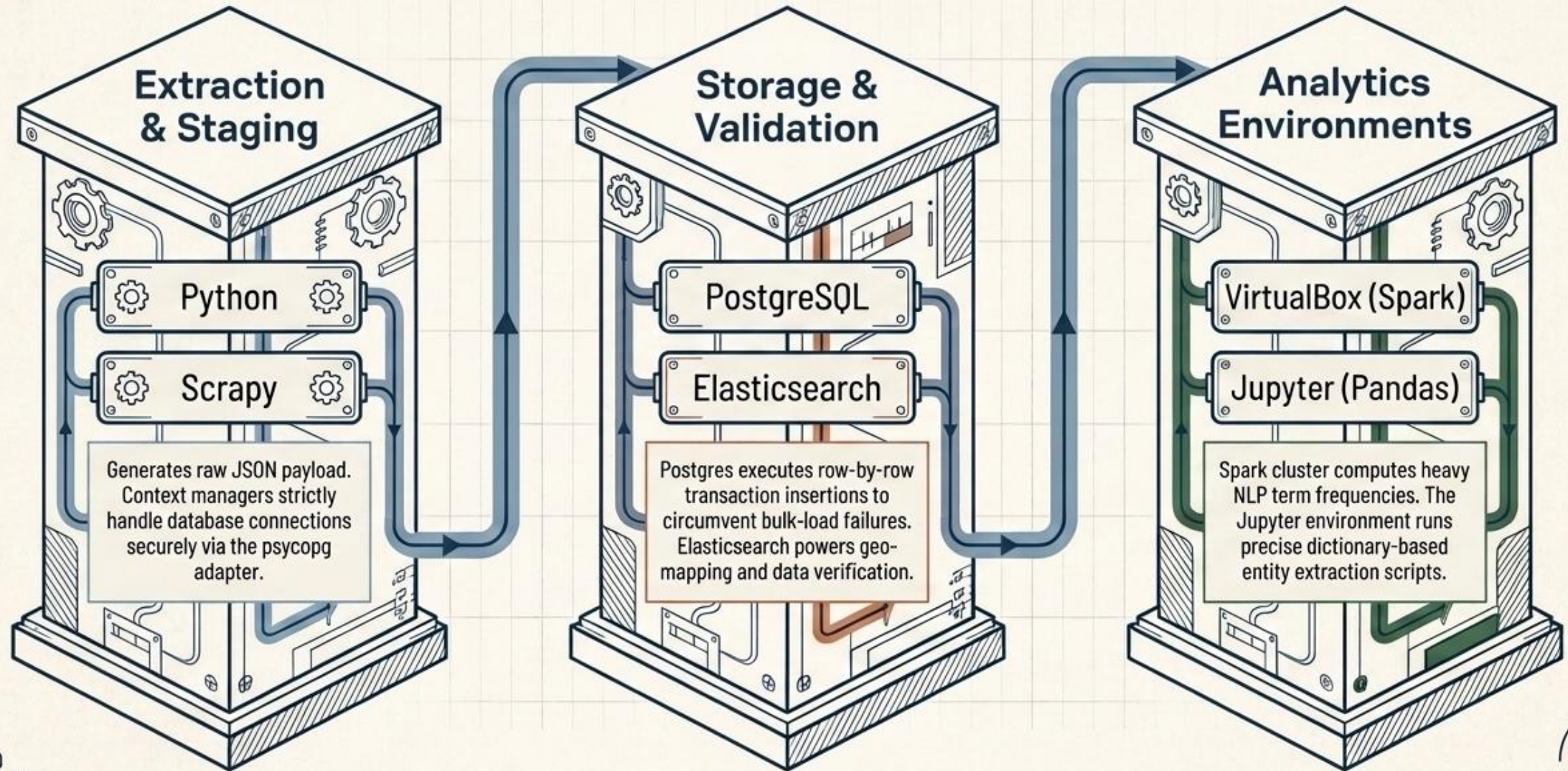
- Develop an end-to-end ETL data pipeline to aggregate Greek IT job postings.
- Extracted data from LinkedIn, Kariera.gr, Skywalker.gr, and Jobfind.gr
- The main part of the dataset is job posts from 10/2025 until 02/2026 and the size is 3184 job posts.
- Handling differences in formatting of some fields like date and location normalisation.

Technical Infrastructure

The project was built upon four technically connected yet distinct components

- **Data Extraction and Storage:** Utilizing the Scrapy framework for crawling and PostgreSQL for relational data storage
- **Elasticsearch and Kibana:** Used for data validation, interactive mapping of job locations and real-time field statistics
- **Spark (VirtualBox):** Employed for Natural Language Processing tasks, including Term Frequency and TF-IDF algorithms
- **Pandas (Jupyter Notebook):** The primary environment for the "Dictionary-based Entity Extraction" method.

Architecture Stack



Methodologies : TF-IDF and Dictionary-based Entity Extraction

- **TF-IDF (Term Frequency-Inverse Document Frequency):** A statistical technique designed to quantify the importance of a word within a specific document relative to the entire corpus. It balances local term frequency with global rarity to identify the unique "lexical footprints" of job roles.
- **Dictionary-based Entity Extraction (SARO Adaptation):** A deterministic approach to Named Entity Recognition that identifies predefined terms through direct string matching. The Skills and Recruitment Ontology (SARO) was adapted into a customized gazetteer, allowing for the extraction of programming languages and databases from unstructured text.

Analysis Engines

Term Frequency-Inverse Document Frequency

Executed via Apache Spark

Goal: Discover relative semantic importance and unique vocabulary traits.

Output: Computationally isolates words uniquely correlated to specific roles.

e.g., Outputting 'big data', 'mining', and 'data engineer' as statistically unique to the 'Data Engineer' job title.

Dictionary-Based Extraction

Executed via Jupyter Notebooks

Goal: Deterministic parsing matching text against predefined taxonomy terms

Output: High-precision identification of hard skills and software tools, deliberately ignoring ambiguous out-of-vocabulary terms.

e.g., Exact matching of 'PostgreSQL' to a structured database lexicon.

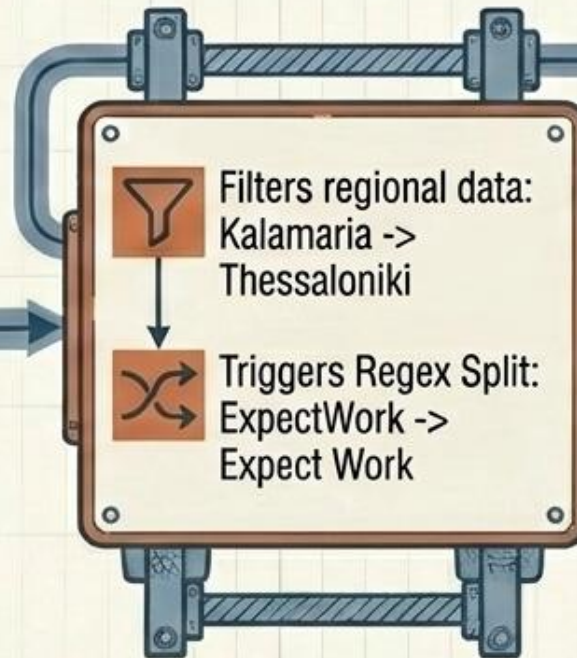
Translating Text to Insight

Station 1 (Input)
Raw Data



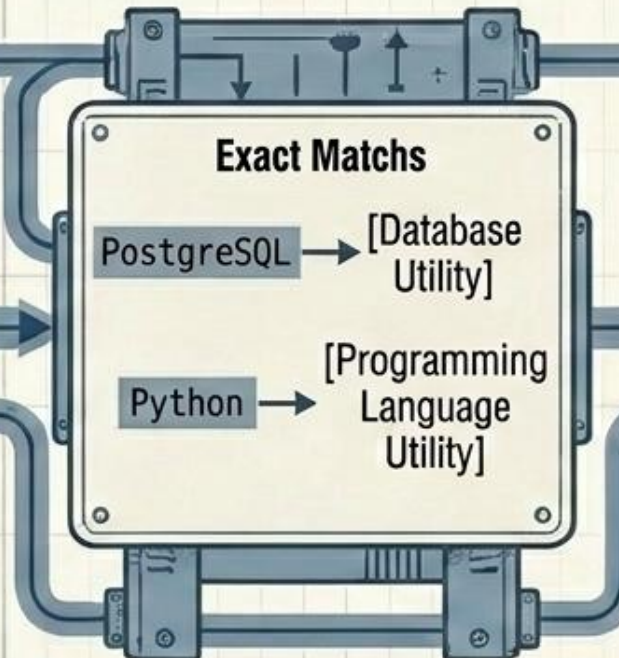
Station 1:
Raw Data

Station 2 (Processing)
The Cleaning Engine



Station 2:
The Cleaning Engine

Station 3 (Matching)
SARO Filters



Station 3:
(Ontology Matching)

Station 4 (Output)
Statistical Increment



Station 4:
Statistical Increment

Methodology – Role Classification & Spatial Investigation

- **Regex-based Role Categorization:**
Utilized regular expression patterns to match job titles against a predefined JSON schema.
- **Multidimensional Field Statistics:**
Integrated Kibana tools to generate statistical summaries and interactive geo-point maps, cross-referencing technical skills with variables like workplace, company, and employment type.

TF-IDF Results

	'Software Engineer'	'Support Engineer'	'Data Engineer'	'IT Support'
0	capacity senior	applicants resolve	capacity data	across greece
1	embedded software	application support	big data	responsibilities
2	software engineer	users collaborate	big	bachelor
3	robotics	teams continue	data engineer	bachelor degree
4	engineer continue	ups	engineer continue	presentation skills
5	τεχνολογίες	internal users	end data	platforms tools
6	senior software	support customers	blockchain	liaison teams
7	saturday	collaborate engineering	data processes	tools additional
8	επαγγελματικής ανάπτυξης	τηλεπικοινωνιών	mining	knowledge learning
9	αντίστοιχη θέση	τμήμα	within data	update training
10	αντίστοιχη	πνεύμα	based experience	users prepare

Dictionary-based Entity Extraction Results

Total number of all mentions of programming languages: 1972			Total number of all mentions of databases: 653		
python	498	25.25%	postgresql	142	21.75%
			sqlserver	135	20.67%
java	313	15.87%	mysql	113	17.30%
			redis	64	9.80%
javascript	298	15.11%	databricks	58	8.88%
			elasticsearch	30	4.59%
go	180	9.13%	snowflake	27	4.13%
			mariadb	22	3.37%
typescript	161	8.16%	bigquery	18	2.76%
			cassandra	12	1.84%
c	144	7.30%	hive	7	1.07%
			oraclesql	6	0.92%
c#	114	5.78%	dynamodb	6	0.92%
			sqlite	5	0.77%
CPlusPlus	92	4.67%	tsql	3	0.46%
			solr	3	0.46%
php	57	2.89%	sql azure	1	0.15%
			mongodb	1	0.15%
kotlin	32	1.62%			
swift	24	1.22%			
scala	21	1.06%			
ruby	20	1.01%			
rust	18	0.91%			

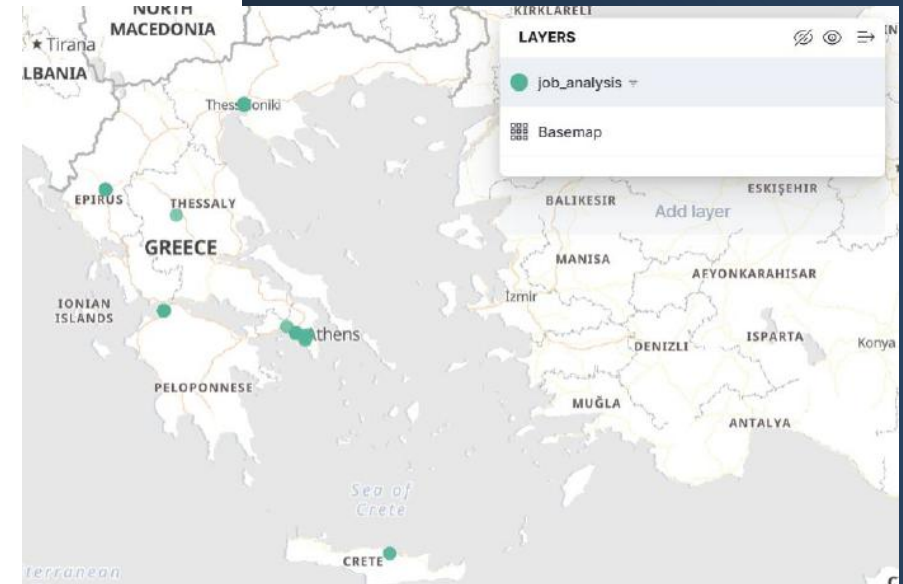
Role Classification of Dictionary-based Entity Extraction Results

Total number of all mentions of programming languages: 1972			--- Demand Distribution for: PYTHON ---		--- Demand Distribution for: JAVA ---	
python	498	25.25%	Data & AI	38.15%	Backend & General Eng	33.55%
java	313	15.87%	Backend & General Eng	22.09%	Management & Architecture	14.7%
javascript	298	15.11%	Infrastructure & Support	15.26%	Full Stack	9.9%
go	180	9.13%	Management & Architecture	10.84%	Data & AI	8.95%
typescript	161	8.16%	Other / Uncategorized	4.22%	Infrastructure & Support	8.63%
c	144	7.30%	Full Stack	4.02%	QA & Testing	7.99%
c#	114	5.78%	QA & Testing	2.81%	Frontend	6.71%
CPlusPlus	92	4.67%	Frontend	2.01%	Other / Uncategorized	4.47%
php	57	2.89%	ERP & Business Systems	0.6%	Mobile	3.19%
kotlin	32	1.62%			ERP & Business Systems	1.92%
swift	24	1.22%				
scala	21	1.06%				
ruby	20	1.01%				
rust	18	0.91%				

Total number of all mentions of programming languages: 1972			--- Demand Distribution for: JAVASCRIPT ---		--- Demand Distribution for: GO ---	
python	498	25.25%	Frontend	34.23%	Backend & General Eng	21.67%
java	313	15.87%	Backend & General Eng	23.15%	Management & Architecture	18.89%
javascript	298	15.11%	Full Stack	17.11%	Data & AI	14.44%
go	180	9.13%	QA & Testing	6.04%	Infrastructure & Support	12.78%
typescript	161	8.16%	Management & Architecture	5.37%	Other / Uncategorized	12.22%
c	144	7.30%	Infrastructure & Support	4.03%	Frontend	8.89%
c#	114	5.78%	Data & AI	3.69%	Full Stack	5.0%
CPlusPlus	92	4.67%	Other / Uncategorized	3.36%	ERP & Business Systems	4.44%
php	57	2.89%	ERP & Business Systems	2.01%	QA & Testing	1.11%
kotlin	32	1.62%	Mobile	1.01%	Mobile	0.56%
swift	24	1.22%	--- Demand Distribution for: TYPESCRIPT ---		--- Demand Distribution for: C ---	
scala	21	1.06%	Frontend	35.4%	Backend & General Eng	28.47%
ruby	20	1.01%	Management & Architecture	20.5%	Infrastructure & Support	23.61%
rust	18	0.91%	Full Stack	16.77%	Other / Uncategorized	12.5%
			Backend & General Eng	16.15%	Data & AI	11.81%
			QA & Testing	6.83%	Management & Architecture	8.33%
			Data & AI	2.48%	QA & Testing	6.25%
			Infrastructure & Support	1.24%	Frontend	3.47%
			Other / Uncategorized	0.62%	Mobile	2.78%
			--- Demand Distribution for: C# ---		--- Demand Distribution for: C++ ---	
			Backend & General Eng	55.26%	Backend & General Eng	56.52%
			Full Stack	9.65%	Data & AI	17.39%
			Management & Architecture	7.89%	Other / Uncategorized	8.7%
			Data & AI	7.89%	Infrastructure & Support	7.61%
			QA & Testing	7.89%	Management & Architecture	6.52%
			Other / Uncategorized	6.14%	Mobile	2.17%
			Frontend	3.51%	Full Stack	1.09%
			Infrastructure & Support	1.75%		
			--- Demand Distribution for: SWIFT ---		--- Demand Distribution for: KOTLIN ---	
			Mobile	54.17%	Mobile	46.88%
			Infrastructure & Support	12.5%	Backend & General Eng	15.62%
			Backend & General Eng	8.33%	Other / Uncategorized	9.38%
			Frontend	8.33%	Frontend	6.25%
			Other / Uncategorized	8.33%	Data & AI	6.25%
			Management & Architecture	4.17%	QA & Testing	6.25%
			QA & Testing	4.17%	Management & Architecture	6.25%
					Infrastructure & Support	3.12%

Python on the Kibana Map

- This is an example of the Elasticsearch map search.
- The same can be done for any text in the dataset.



Results: Roles & Regional Variation

Significant Findings & Key Insights

- **Python leads the market (25.3%)** as the primary language for Data/AI and Backend, while **Java (15.9%)** maintains a strong enterprise presence in the Greek ICT sector.
- **PostgreSQL (21.8%)** is the preferred database, with **Microsoft SQL Server (20.7%)** showing a notably higher regional demand in Greece compared to global averages.
- **Geographic Concentration:** The vast majority of IT roles are heavily centralized in Athens, followed by Thessaloniki.
- **Methodology Efficiency:** The Dictionary-based Entity Extraction identified specific programming languages and roles with higher accuracy than pure TF-IDF.
- **Market Alignment:** Results mostly align with the 2025 Stack Overflow global trends, identifying a "Modern Tech Stack" preference.

Conclusions & Future Work

Conclusions

- Aggregating unstructured real time job portal data can generate a high-level image of a sector.
- This is technologically feasible and potentially scalable.

Future Work

- If implemented at scale, these methods could be applied not only to the ICT sector but in the whole labor market, accurately depicting the demand of specific professions, their location and their requirements contents.
- Create AI agents within Elasticsearch customized for the dataset enabling users to query the dataset directly using natural language.
- Transitioning from Regex/TF-IDF to advanced Machine Learning for higher accuracy.



Thank you for your attention.

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